

NITHIN V

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Integrated M.Sc. candidate in Artificial Intelligence & Machine Learning with hands-on experience building AI solutions across healthcare, finance, and document intelligence. Proficient in Python, TensorFlow, FastAPI, and Docker, with exposure to LLMs, RAG, and Computer Vision.

EDUCATION

Integrated Masters in Artificial Intelligence & Machine Learning

July 2022 – May 2027

Coimbatore Institute of Technology, Coimbatore

Cgpa 8.31 / 10 (VIII Sem)

EXPERIENCE

AI/ML Intern | TealOrca Software Solutions Pvt. Ltd.

June 2025 – November 2025

- **70–80% of 100 daily calls** resolved autonomously by multilingual voice AI agents (Tamil & English) built for appointment booking, batch scheduling, and customer enquiries — remainder routed via live transfer.
- **5–7 domain-specific voice agents** (healthcare, finance, restaurant) deployed live over phone using **ElevenLabs + Twilio** — human-like conversations engineered through structured multilingual prompt design.
- **All 9 Voice API modules** (Agents, Batches, Phone Calls, Knowledgebases, Inbound Agent, Executions, Phone Numbers, User, Voice) developed in Python + FastAPI and **fully shipped to production** within internship timeline.

PROJECTS

FinAgent – Explainable Financial Risk & Compliance Intelligence Platform

[Source Code](#)

- Designed and developed an explainable financial intelligence platform for **bankruptcy risk assessment, transaction fraud detection, payment fraud simulation** using the Polish Bankruptcy, IEEE-CIS Fraud Detection, and PaySim datasets; built XGBoost-based predictive models, engineered domain-specific financial and transaction features, and integrated **SHAP** explainability to identify the key drivers behind model decisions.
- Built a deterministic **compliance and risk-governance engine** integrating model predictions, risk flags, and recommendations; developed a **FastAPI + React** platform with PDF export and end-to-end validation.

Orcan Vision Trace – Reverse Image Search

[Source Code](#)

- **90–95% retrieval accuracy (AWS Rekognition) vs 60% (custom FAISS)** across **50,000+ facial images** - GPU-based FAISS indexing completes in 20 minutes with zero cloud dependency.
- **60% cost reduction per query**- Consumer-facing AWS Rekognition mode was priced at \$0.03–0.05 per image in the app — local GPU-based FAISS pipeline brought this down to \$0.02, offering clients a cost-effective solution.

Videobot – Video Interactive Chat Assistant

[Source Code](#)

- **Exact timestamp range retrieval in 5 seconds**: The RAG pipeline chunks 20–30 min educational lectures into ChromaDB – chat queries trigger video playback to the precise start-to-end segment containing the answer.
- **Multimodal pipeline** combining audio extraction, subtitle generation, vector chunking (ChromaDB), and LLM response — synchronized chat + video UI delivers document-faithful answers with zero hallucination drift.

TECHNICAL SKILLS

Languages: Python, SQL, React JS

Frameworks & Libraries: NumPy, Pandas, TensorFlow, PyTorch, Matplotlib, Scikit-learn, Streamlit, FastAPI

AI/LLM Techniques: Retrieval-Augmented Generation (RAG), Vector Databases, Computer Vision, Conversational AI,

Tools: Git & GitHub, Docker, PostgreSQL, Power BI, n8n, ElevenLabs, Google Cloud - Vertex AI, AWS

Data Techniques: Data Cleaning, Exploratory Data Analysis (EDA), Data Interpretation, Data Visualization

Machine Learning Concepts: Supervised and Ensemble Learning, Artificial Neural Networks, Deep Learning

ACHIEVEMENTS & CERTIFICATIONS

- [AI and ML for Geodata Analytics](#), ISRO and the Indian Institute of Remote Sensing (IIRS), Dehradun.
- [Learning Analytical Tools](#), NPTEL.
- [LifeTrx HealthTech Hackathon](#), Ranked among the top 50 out of 800+ participants for HealthTech innovation.
- [GDG's Hackathon](#), Ranked among the top 25 out of 111+ participants for AI-driven Ear Health Analytics.
- [Event Organizer, AIWO Influence 2025](#), AIWO.
- [Student Coordinator, Institution's Innovation Council \(IIC\)](#), Coimbatore Institute of Technology.